Lec 41

Gaussian vectors

X is a zero mean Gaussian vector if $\exists$ a matrix A st

$$\underbrace{X}_{n\times 1} = \underbrace{A}_{n\times m}\underbrace{W}_{m\times 1} \quad \text{where} \quad W = \begin{pmatrix} W_1 \\ \vdots \\ W_m \end{pmatrix} \quad W_i \sim N(0,1) \text{ and are iid.}$$

$K_X = E[\bar{X}\bar{X}^T]$ is p.s.d matrix.

Covariance matrix

$$\Updownarrow$$

$$\exists A \text{ st } K_X = AA^T$$

If K_X is non-singular (invertible)

$$f_X(\underline{x}) = \frac{1}{(2\pi)^{n/2}|K_X|^{1/2}} e^{-\frac{1}{2}\underline{x}^T K_X^{-1}\underline{x}}$$

What if K_X is singular?

$\rightarrow$ can get samples of X from W by linear transformation $X = AW$

Examples:

$$K_X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$K_X = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow A = \begin{bmatrix} \sqrt{4} & 0 \\ 0 & \text{[illegible]} \end{bmatrix}$$

$$K_X = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} \rightarrow A = \underbrace{\begin{bmatrix} \; \end{bmatrix}}_{Q}\underbrace{\begin{bmatrix} \; \end{bmatrix}}_{D^{1/2}}\underbrace{\begin{bmatrix} \; \end{bmatrix}}_{Q^T}$$

samples of X
by linear
= AW

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow A = \begin{bmatrix} \sqrt{4} & 0 \\ 0 & \end{bmatrix}$

$\Rightarrow A = \underbrace{\begin{bmatrix} & \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} & \end{bmatrix}}_{D^{1/2}} \underbrace{\begin{bmatrix} & \end{bmatrix}}_{Q^T}$

Assume $n=2$

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

$$K_X = \begin{pmatrix} \sigma_{x_1}^2 & \rho\sigma_{x_1}\sigma_{x_2} \\ \rho\sigma_{x_1}\sigma_{x_2} & \sigma_{x_2}^2 \end{pmatrix}$$

$$\rho = \rho(X_1, X_2) = \frac{Cov(X_1, X_2)}{\sqrt{Var(X_1)}\sqrt{Var(X_2)}}$$

$$K_X = QDQ^T$$

$$\boxed{A = QD^{1/2}Q^T}$$

$QD^{1/2}UU^TD^{1/2}Q^T$

$\boxed{A = QD^{1/2}U}$

$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi}$

$f_{X_1,X_2}(x_1,x_2) =$

(1) What is K_X is

$\exists$ a matrix $K_X =$

A st

A

Wlog let

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sqrt{(1-\rho^2)}\,\sigma_{x_1}\sigma_{x_2}} e^{-\frac{1}{2}\left[\left(\frac{x_1}{\sigma_{x_1}}\right)^2+\left(\frac{x_2}{\sigma_{x_2}}\right)^2-2\frac{\rho x_1 x_2}{\sigma_{x_1}\sigma_{x_2}}\right]\left(\frac{1}{1-\rho^2}\right)}$$

$$\Updownarrow$$

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{(2\pi)^{2/2}|K_X|^{1/2}} e^{-\frac{1}{2}(x_1\ x_2)K_X^{-1}\begin{pmatrix}x_1\\x_2\end{pmatrix}}$$

$\rho\sigma_{x_1}\sigma_{x_2}$

σ_{x_2} $\sigma_{x_2}^2$

$X_2)$

$x_1, x_2]$

$(x_1)\sqrt{var(x_2)}$

$D Q^T$ | $Q D^{1/2} U U^T D^{1/2} Q^T$

$D^{1/2} Q^T$ | $\boxed{A = Q D^{1/2} U}$

① What is K_X is not invertible?

$\exists$ a matrix A st $K_X = AA^T$ $X = AW$

A is also a singular matrix.

Wlog let first l rows of A be li

Let them be called $\hat{A}$

$K_X = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ $\lambda_1 = 5$ $\lambda_2 = 0$

Sum of eigen values = trace of matrix = 5

Product of eigenvalues = $\det(K_X) = 0$

$A = \frac{1}{\sqrt{5}}\begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$AA^T = \frac{1}{5}$

$= \frac{1}{5}$

$$A = \frac{1}{\sqrt{5}} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

can check that $AA^T = K_X$

$$\boxed{R = Q D^{1/2} Q^T}$$

Check

$$A A^T = \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} = K_X$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$$

$$X_1 = \frac{1}{\sqrt{2}} (4W_1 + 2W_2)$$

$$X_2 = \frac{1}{\sqrt{2}} (2W_1 + W_2) = \frac{X_1}{2}$$

$$X = \begin{pmatrix} \hat{X} \\ \overline{X} \end{pmatrix}$$

$$\hat{X} = \underset{r \times n}{\hat{A}} W$$

$$f_X(x) = f_{\hat{X}}(\hat{x}) \delta(x_{r+1} - \ ) \delta(x_{r+2} - \ ) \cdots \delta(x_n - \ )$$

$$K_{\hat{X}} = \hat{A} \hat{A}^T$$

$$f_{\hat{X}}(\hat{x}) = \frac{e^{-\frac{1}{2} \hat{x}^T K_{\hat{X}}^{-1} \hat{x}}}{(2\pi)^{r/2} |K_{\hat{X}}|^{1/2}}$$

$$\boxed{f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) \, \delta\left(x_2 - \frac{x_1}{2}\right)}$$

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Assume $n=2$.

We want to find points such that the pdf at these points evaluates to 80% of the maximum possible value.

Find all $\underline{x} = (x_1, x_2)$ st. $f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi |K_X|^{1/2}} e^{-\frac{1}{2}(x_1\ x_2) K_X^{-1} \binom{x_1}{x_2}}$

$$= \frac{80}{100} \times \frac{1}{2\pi |K_X|^{1/2}}$$

$$\Rightarrow \boxed{e^{-\frac{1}{2}\underline{x}^T K_X^{-1} \underline{x}} = \frac{80}{100}}$$

X is zero-mean Gaussian vector with K_X. Covariance matrix

$$f_X(\underline{x}) = \frac{e^{-\underline{x}^T K_X^{-1} \underline{x}}}{(2\pi)^{n/2} |K_X|^{1/2}}$$

$X \xrightarrow{g(x)} U \sim N(\underline{\mu}, K_X)$

$f_U(u) = \frac{f_X(g^{-1}(u))}{g'(x)}$ $\quad U \overset{g(x)}{=} X + \underline{\mu}$

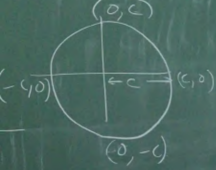
$$f_U(\underline{u}) = f_X(u-\mu) = \frac{e^{-(u-\mu)^T K_X^{-1}(u-\mu)}}{(2\pi)^{n/2}|K_X|^{1/2}}$$

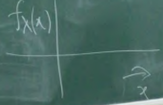
$$\underline{x}^T K_x^{-1} \underline{x} = \boxed{2 \ln \frac{100}{80}} = c$$

$$(x_1 \ x_2) K_x^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = c$$

Examples: $K_x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow x_1^2 + x_2^2 = c$$

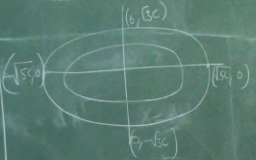


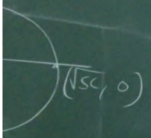


$$K_x = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

$$(x_1 \ x_2) \begin{pmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = c$$

$$\Rightarrow \boxed{\frac{x_1^2}{5c} + \frac{x_2^2}{3c} = 1}$$





$$K_x = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \qquad D = \begin{bmatrix} 3/2 & \\ & 1/2 \end{bmatrix}$$

$$K_x = Q\boxed{D}Q^T$$

$$K_x^{-1} = (Q^T)^{-1} D^{-1} Q^{-1} = Q D^{-1} Q^T$$

$$\underline{x}^T Q D^{-1} Q^T \underline{x} = c$$

$$Q = \begin{bmatrix} q_{11} & q_{21} \\ q_{21} & q_{22} \end{bmatrix}$$

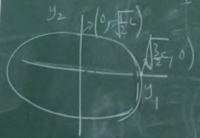
$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Let $\underline{y} = Q^T \underline{x}$ $\qquad \underline{x} = Q\underline{y}$

then $\underline{y}^T D^{-1} \underline{y} = c$

$$\Rightarrow \frac{y_1^2}{\frac{3c}{2}} + \frac{y_2^2}{\frac{1c}{2}} = 1$$

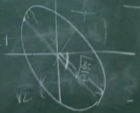
$$y = \begin{pmatrix} 1 & -1 \\ & \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\theta = -45^\circ$$

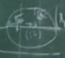
$$\frac{(x_1 - x_2)^2}{2\cdot\frac{3c}{2}} + \frac{(x_1 + x_2)^2}{2(\frac{1}{2}c)} = 1$$



$$[y_1\ y_2]\begin{pmatrix} \frac{1}{3c/2} & \\ & \frac{1}{c/2} \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Suppose $\underline{x}$ has mean $\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$

Example: $\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $K = \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix}$



$$(x_1 - \mu_1 \ \ x_2 - \mu_2)\begin{pmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}\begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix} = c$$

$$\frac{(x_1 - \mu_1)^2}{5c} + \frac{(x_2 - \mu_2)^2}{3c} = 1$$

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$$\begin{pmatrix} x_1-\mu_1 & x_2-\mu_2 \end{pmatrix} Q D^{-1} Q^T \begin{pmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{pmatrix} = c.$$

$$K_x = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

to

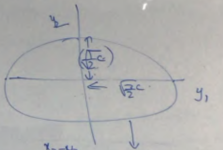
$$Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 3/2 & \\ & 1/2 \end{bmatrix}$$

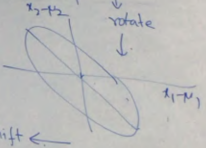
$$\underline{y} = Q^T \begin{pmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{pmatrix}.$$

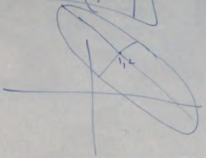
$$\frac{y_1^2}{c\lambda_1^2} + \frac{y_2^2}{c\lambda_2^2} = 1.$$

$$\boxed{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = Q\,\underline{y} + \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}}$$

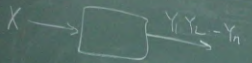


rotate





Shift



$$P(X=x \mid Y_1 Y_2 \dots Y_n = y_1 \dots y_n)$$

$$= \frac{P(Y_n=y_n \; X=x \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}{P(Y_n=y_n \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}$$

$$= \frac{P(Y_n=y_n \mid X=x, Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1}) \; P(X=x \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}{P(Y_n=y_n \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}$$

$$= \frac{P(Y_n=y_n \mid X=x) \; P(X=x \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}{\sum_{x} P(Y_n=y_n \mid X=x) \; P(X=x \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})}$$

Finish!

assuming this as prior update can be done

Past updates helped you find

$$P(X=x \mid Y_1 \dots Y_{n-1} = y_1 \dots y_{n-1})$$

$$P(X=x)$$
$$\downarrow$$
$$P(X=x \mid Y_1=y_1)$$
$$\downarrow$$
$$P(X=x \mid Y_1=y_1, Y_2=y_2)$$

Random Processes

Sequence of random variables $(X_n)_{n \in \mathbb{N}}$ or $(X(t))_{t \in \mathbb{R}}$ are refered to as a random processes.

Discrete time random process : $(X_n)_{n \in \mathbb{N}}$.
countably many random variables

Random variables themselves can be continuos or discrete.

Continuos time random process : $(X(t))_{t \in \mathbb{R}}$.
uncountable # of r.vs.

Discrete time random process

<u>Example</u> :

1. Markov chains $(X_1, X_2, X_3 \ldots)$
(discrete time)

$$P(X_t = x_t \mid X_1 = x_1 \ldots X_{t-1} = x_{t-1})$$

$$= P(X_t = x_t \mid X_{t-1} = x_{t-1}).$$

2. Bernoulli process.

$X_1, X_2 \ldots$ $X_i \sim \text{Bernoulli}(p)$.
and iid.

# of Successes / Counting process $\left\{ \begin{array}{l} N_1 \; N_2 \ldots \\ N_n = \sum_{i=1}^{n} X_i \end{array} \right.$

Time between two successes $\leftarrow$ $T_1 \; T_2 \; T_3 \ldots$

Can use any one process below to describe the rest

$X_1 \; X_2 \; \ldots$
$T_1 \; T_2 \; \ldots$
$N_1 \; N_2 \; \ldots$
$S_1 \; S_2 \; \ldots$

$$T_1 = \min\{n : X_n = 1\}.$$

$$T_2 = \min\{n - T_1 : n > T_1, X_n = 1\}.$$

$$T_i = \min\{n - T_{i-1} : n > T_{i-1}, X_n = 1\}$$

$$S_1 \; S_2 \; S_3 \ldots \quad S_n = \sum_{i=1}^{n} T_i$$

③ Auto-regressive process

$$X_1, X_2, \ldots$$

$$X_t = c + \phi X_{t-1} + \varepsilon_t.$$

where $\varepsilon_t \sim N(0, \sigma_t^2)$. ε_t's are independent.

$X_0 = 0$, c, ϕ constants

Here the RVs X_i's are continuos.

Used to model stock prize changes, weather updates correlated with previous day with some added randomness

Continuos time random process.

① Poisson process.

Discrete time process $\begin{cases} T_1 \quad T_2 \quad T_3 \quad \cdots \\ T_i \sim \text{exponential}(\lambda). \text{ and iid.} \end{cases}$

\# of packets arrived before time t $\begin{cases} N(t) = \min\{n : \sum_{i=1}^{n} T_i \leq t\}. \end{cases}$

$$S_n = \sum_{i=1}^{n} T_i .$$

Can use any one of them to describe rest

$T_1 \quad T_2 \quad \cdots$

$S_1 \quad S_2 \quad S_3 \quad \cdots$

$(N(t))_{t \in \mathbb{R}}.$

Can show that $N(t) \sim \text{Poisson}(\lambda t)$.